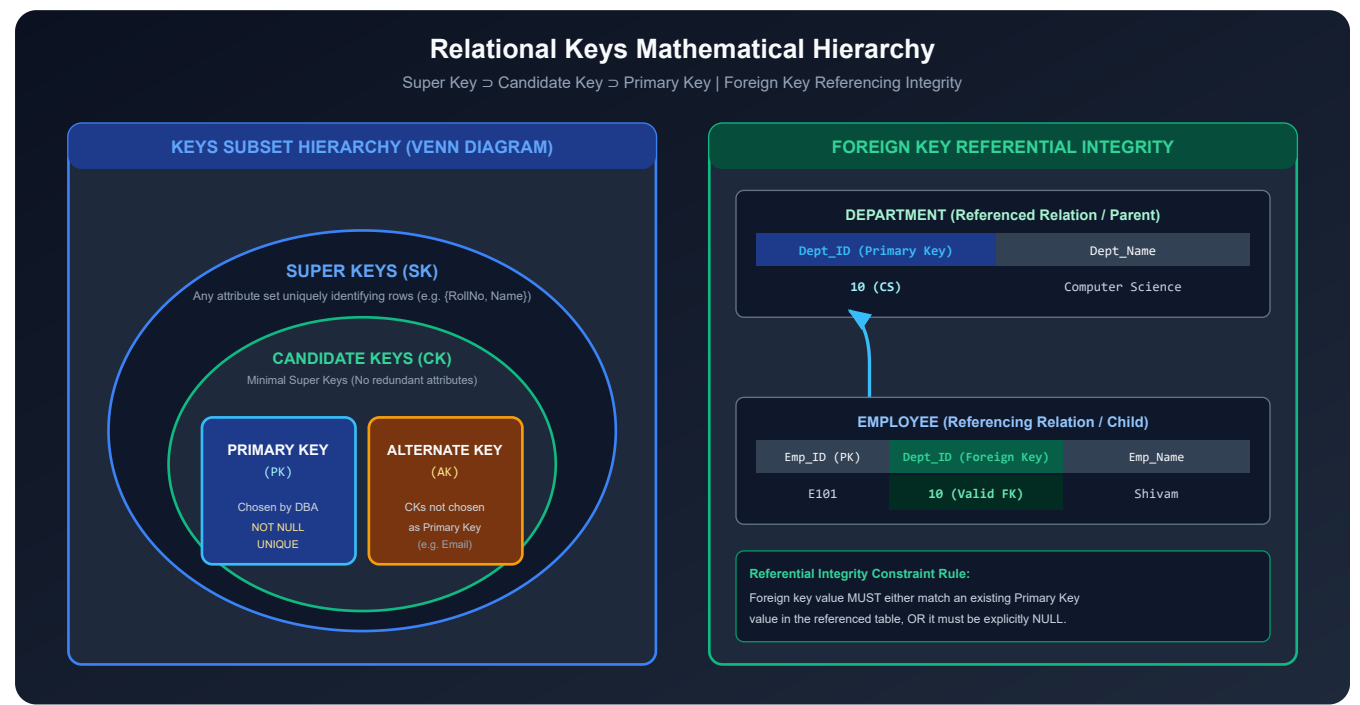


Module 07: Relational Keys in Depth

1. Keys Hierarchy & Foreign Key Architecture



2. Comprehensive Key Types

Key Type	Minimality	NULL Permitted?	Purpose
Super Key	No (Can contain extra attributes)	Yes	Uniquely identifies tuples
Candidate Key	Strictly Minimal	No (usually)	Eligible candidate to become PK
Primary Key	Strictly Minimal	Strictly NEVER NULL	Central unique row identifier
Alternate Key	Strictly Minimal	Yes	Candidate keys not picked as PK
Foreign Key	Depends on target PK	YES (Unless NOT NULL constraint)	Cross-table referential integrity

3. Solved Super Key Formula

If a relation has n attributes and a candidate key of size k attributes, the total number of super keys containing this candidate key is:

Total Super Keys = $2^{(n - k)}$